

AT-FSV / ATAI — CMFSB ISOLATION MODULE

ATFSV-MOD-001 | Captured Mechanical Floating Safety Body

DOCUMENT 00 — MASTER ENGINEERING BASELINE

Revision 4.7 | Module-Primary External Baseline | Dual-Mode Comfort / Sport | 2026-10-02

Founder / Legal Owner: Amirbahador Atai

Module Datasheet Summary (ATFSV-MOD-001)

Parameter	Value	Status
Architecture	Mechanically captured upper body + macro–micro isolation (NOT free levitation)	BASELINE
Dual operating modes	SPORT (rigid lock / handling) + COMFORT (soft isolation) — see §22	BASELINE
Mode selection paths	Driver command / radar-based / GPS–map-based (three-path arbitration)	TARGET
Reference topology	Hybrid hydraulic (macro) + Lorentz (micro)	BASELINE path
PM+EM reluctance	OPTION only — stability gate; 15–30 mm gap applies to OPTION only	OPTION
Static load classes	Size S/M/L: see §5 and §13	STUDY
Macro stroke	±50 mm body-relative (100 mm total range)	TARGET
Micro travel (Lorentz)	±7.5 mm stroke; coil–magnet gap few mm (ICD)	TARGET
Peak dynamic force	±2.5 kN per tower (study); higher after proof	STUDY
Continuous RMS force	≈0.7 kN RMS per tower (aligned with ≤0.5 kW mean for 4 towers)	STUDY
Motor constant Km	≥63 N/√W ASSUMPTION — confirm by FEA / supplier	ASSUMPTION
Mean copper-loss power (4 towers)	≤0.5 kW continuous class at 0.7 kN RMS	TARGET
Peak pulse (4×2.5 kN)	≈6.3 kW class; ≈131 A at 48 V — ICD sizing	STUDY
Module supply	48 V class nominal (final ICD)	STUDY
Control bandwidth (micro)	≥20 Hz TARGET; carriage first mode ≥3× bandwidth	TARGET
Module mass boundary	Core tower ≤55 kg TARGET; full corner STUDY — see §5	TARGET / STUDY
Leveling	Active; fail-closed valves on power loss	TARGET
Body lock (SPORT)	Mechanical / hydraulic rigid engagement; upper body seats on lower body — see §22	TARGET
Power-loss behaviour	Passive mechanical/hydraulic support within envelope (S4)	BASELINE invariant
Life	≥10,000 h equivalent road duty TARGET	TARGET
Primary commercial path	Path A — module field license / supply; Option framework §23	—
Lead application	Reference Application A — see §13; dual-mode applicable across classes	PROPOSED
Status	PRELIMINARY — NOT CERTIFIED — TRL 2–3	—

STATUS: Preliminary Engineering Baseline — Not Certified. No test result, certification, survivability demonstration or patentability is asserted. Values marked TARGET, STUDY, OPEN or ASSUMPTION are not compliance claims. Vehicle-level mass, powertrain and range data are in Appendix A (reference vehicle only). Founder / Rights holder: Amirbahador Atai.

0. Document Control and Status Vocabulary

Field	Value
Document ID / Revision / Date	ATFSV-DOC-00-R4.7 / 4.7 / 2026-10-02
Supersedes	ATFSV-DOC-00-R4.6
Legal Owner / Rights holder / Founder	Amirbahador Atai
Document Owner	ATFSV Project Engineering Documentation Authority
Author	ATFSV Engineering
Technical Reviewer	Independent engineering review required before design use — to be appointed
Approver	Not approved for road use, production or certification credit — CONTROLLED
Distribution	Controlled — NDA, due-diligence and supplier engagement only
Classification	Confidential — IP / Engineering Development
Configuration status	NOT configuration-controlled until formal CM system established
Purpose	Module architecture baseline, dual-mode Comfort/Sport, mass/energy analysis, safety planning, supplier engagement, NDA due diligence

Chain of title and disclosure logs are maintained outside this document. External release follows the tiered scheme in §16. This document is not legal advice.

0.1 Status Vocabulary (exclusive)

Label	Mandatory meaning
CONTROLLED DESIGN LIMIT	Architecture ceiling; not a measured result.
BASELINE	Included in current architecture accounting.
TARGET	Numerical target requiring verification.
STUDY	Study allocation or scenario; not frozen design.
ASSUMPTION	Modelling assumption needing confirmation.
DERIVED	Calculated from stated assumptions.
OPTION	Not credited in baseline safety or mass closure.
OPEN	Unresolved; owner and gate required.
TBD	Not responsibly fixed at this revision.
VERIFIED	Reserved for objective CM evidence only — not used in R4.7.

Rule: TARGET, STUDY, OPEN and ASSUMPTION are not compliance claims. A value becomes a controlled design requirement only after requirement ID, owner, verification method and evidence reference are baselined.

0.2 Value Statement (screening)

Problem. Occupant compartments in specialty, defence-utility, ambulance and luxury extreme vehicles receive road and terrain vibration that degrades comfort, clinical stability, task performance and long-duration fatigue. Conventional suspensions couple road input into the body across ISO 2631-1 weighted bands and the motion-sickness band (approximately 0.1–0.5 Hz). Soft isolation alone can degrade handling in rapid manoeuvres (e.g. moose test).

Solution concept. CMFSB keeps the upper body mechanically captured at all times. Macro travel (± 50 mm body-relative, 100 mm total range) provides large-stroke isolation on a passive support path with active leveling. Micro travel (± 7.5 mm) provides high-bandwidth force control via the Lorentz stage. Static load is never credited to magnetic force. R4.7 adds explicit dual operating modes — **SPORT** (upper body seats and locks to lower body for handling) and **COMFORT** (upper body isolates for soft ride) — with three selection paths (driver, radar, GPS/map). “Floating” means body isolation relative to the chassis — not free levitation and not marine flotation.

Criterion	R4.7 statement	Status
Ride isolation (COMFORT)	Transmissibility vs defined passive hydro-pneumatic reference in 0.1–0.5 / 0.5–2 / 4–12.5 Hz	OPEN — ATFSV-SIM-001
Handling (SPORT)	Upper body locked/seated; effective roll/yaw stiffness approaches rigid-body class	TARGET — §22
Mode transition time	≤ 3 s typical TARGET (full SPORT $\leftrightarrow$ COMFORT under defined conditions)	TARGET
Module mass (core tower)	≤ 55 kg TARGET (structure + Lorentz + sensors + local electronics)	TARGET
Dynamic force (peak)	± 2.5 kN study per module	STUDY
Dynamic force (RMS continuous)	≈ 0.7 kN RMS planning (aligned with ≤ 0.5 kW mean copper loss for 4 towers)	STUDY
Static support	Passive path sized for occupied and size-class states	STUDY
Power-loss behaviour	Defaults to passive support within envelope; leveling valves fail-closed	BASELINE invariant
Leveling	Active; preserves dynamic stroke under payload change	TARGET

Target markets (PROPOSED): Specialty & defence utility; luxury extreme / limited-series vehicles; Reference Application A variants. **Path A** = module field license (primary). **Path B** = vehicle partnership (Appendix A). No comfort percentage is claimed until simulation or test exists.

1. Engineering Baseline and Controlled Values

R4.7 defines a mechanically captured isolation architecture with explicit dual operating modes (SPORT / COMFORT). Static load is carried by compliant mechanical / hydraulic support; Lorentz actuation is a bounded dynamic-control layer. Module parameters are primary. Vehicle-level mass, powertrain, range and brake figures are retained in **Appendix A (Reference Vehicle)** for context only and are not prerequisites for Path A module licensing.

Parameter	R4.7 value	Status
Upper-body empty mass (ref. vehicle)	2,150–2,350 kg P50/P90 study band	STUDY — App A
Upper-body occupied (4 $\times$ 100 kg)	$\approx 2,550$ – $2,750$ kg	DERIVED — App A
Upper-body maximum study	$\approx 2,950$ – $3,150$ kg	STUDY — App A
Ref. Application A cabin mass	1,500–2,500 kg class (confirm empty vs equipped)	STUDY
Size classes (static load / tower)	S ≈ 3 –5 kN; M ≈ 5 –7 kN; L ≈ 7 –8 kN (screening)	STUDY
Reference topology	Hybrid hydraulic macro + Lorentz micro	BASELINE path
PM+EM reluctance	OPTION — stability gate; 15–30 mm working gap applies to OPTION only	OPTION
PM bias ceiling (OPTION only)	≤ 1.5 kN per active module nominal	CONTROLLED DESIGN LIMIT
Lorentz coil–magnet gap / stroke	Few mm gap (ICD); ± 7.5 mm stroke	TARGET
Macro compliant travel	± 50 mm body-relative (100 mm total range)	TARGET
Passive / rate-limited hold	Optional; hydraulic orifice or friction; not primary crash path — see §6	BASELINE
SPORT body lock	Rigid engagement of upper to lower body; see §22	TARGET
COMFORT isolation	Full macro–micro isolation envelope active; see §22	TARGET

2. System Architecture and Safety Invariants

Layers (top $\rightarrow$ bottom): Upper occupant/safety body $\rightarrow$ Micro stage (local carriage + Lorentz) $\rightarrow$ Macro stage (hydro-pneumatic + active leveling + lock interfaces) $\rightarrow$ Lower vehicle body/chassis. Separation is functional and structural — not a claim of levitation. In SPORT mode the upper body is seated and locked to the lower structure; in COMFORT mode the isolation stages operate within their design envelopes.

Figure 1 — Layer stack (conceptual)

Layer	Function	Travel / notes
UPPER OCCUPANT / SAFETY BODY	Occupant compartment; mechanically captured	Mass states / size classes
MICRO STAGE (carriage + Lorentz)	High-bandwidth force; coil–magnet gap few mm	± 7.5 mm stroke; peak ± 2.5 kN study
MACRO STAGE (hydro-pneumatic + leveling + lock)	Primary ride isolation; static load; leveling; SPORT lock seats	± 50 mm body-relative; fail-closed valves; lock faces
LOWER VEHICLE BODY / CHASSIS	Structure, propulsion, brakes, hard-points, lock receivers	Capture frame and progressive stops

Magnetic subsystem is never the sole vertical load path. Progressive stops remain the ultimate crash path. SPORT lock is secondary to capture and stops for crash energy.

Safety invariant	R4.7 rule
Mechanical capture	No operating state permits loss of geometric capture.
Static support	Vehicle weight supported by passive or fail-safe compliant load paths.
Magnetic control	Lorentz force is bounded and cannot be the sole vertical load path.

Safety invariant	R4.7 rule
Loss of electrical power	Transitions to mechanically stable configuration without software aliveness; leveling valves fail-closed.
Crash load path	Progressive stops and structural members carry crash loads without closed magnetic control loop.
Passive / rate-limited hold	Not a primary crash arrestor. Engagement rate-limited; stops remain ultimate load path.
Leveling	Active leveling preserves available dynamic stroke under payload change; failure modes in FMEA.
Mode transition integrity	SPORT↔COMFORT transitions never enter free-floating without mechanical/hydraulic constraint; see §22.
SPORT lock under fault	Lock failure shall not disable capture or progressive stops; vehicle remains controllable in degraded isolation.
Human authority	Driver and rescue retain physical access and defined manual recovery provisions.

3. Macro–Micro Travel, Stiffness Stages and Leveling

R4.7 adopts a macro–micro kinematic architecture. Body-support stiffness is separated from local magnet-carriage guidance. For the Lorentz reference path, carriage guidance is specified by stiffness/clearance and first structural mode $\geq 3\times$ control bandwidth — not by the 133–200 kN/m figures that originated in OPTION PM gap-stiffness formulae. In SPORT mode, macro travel is driven to the seated position and locks engage; in COMFORT mode the full ± 50 mm envelope is available.

Stage	Role	Indicative sizing	Status
Macro body support	Primary ride isolation + static load (COMFORT)	$\approx 1.0\text{--}1.4$ Hz $\rightarrow$ order 25–45 kN/m per corner	OPEN sim
Active leveling	Preserve dynamic stroke under payload change	Hydraulic/air; fail-closed valves on power loss	TARGET
SPORT lock seats	Rigid upper–lower engagement for handling	Face contact + positive lock; see §22.3	TARGET
Local magnet carriage (Lorentz)	Actuator mounting / guidance	First mode $\geq 3\times$ bandwidth; clearance per ICD	STUDY
Micro control (Lorentz)	High-bandwidth force (primarily COMFORT)	± 7.5 mm stroke; ≥ 20 Hz; peak ± 2.5 kN study	TARGET
Progressive stop	Energy limit after excursion	Force–displacement from design/test	OPEN

3.1 Frequency, deflection and leveling rationale

Corner mass share (occupied $\approx 2,650$ kg / 4) ≈ 662.5 kg. At $k \approx 40$ kN/m, $f_n \approx 1.23$ Hz. Payload change $\Delta F \approx 2.4$ kN (empty 5.3 kN $\rightarrow$ maximum 7.7 kN) at $k \approx 40$ kN/m produces ≈ 60 mm differential static displacement — about 60% of the 100 mm total range (± 50 mm). Without leveling, remaining dynamic stroke is only about ± 20 mm, which is insufficient for isolation targets. Active leveling restores the operating point and preserves available dynamic isolation stroke. Air-spring progressive rate partially mitigates but does not eliminate the need for controlled leveling.

Quantity	Value
Corner mass (empty / occupied)	562.5 kg / ≈ 662.5 kg
f_n at $k \approx 32$ / 40 kN/m	≈ 1.2 Hz / ≈ 1.23 Hz — body-support study class
ΔF empty $\rightarrow$ maximum per module	≈ 2.4 kN $\rightarrow \approx 60$ mm at 40 kN/m ($\approx 60\%$ of 100 mm total range)
Remaining dynamic stroke without leveling	$\approx \pm 20$ mm — insufficient for isolation targets
Force authority 4×2.5 / 4×5 kN on 2,650 kg	$\approx \pm 0.38$ g / $\approx \pm 0.77$ g class study

ATFSV-SIM-001 shall report transmissibility for body-support stiffness $\approx 1.0\text{--}1.4$ Hz and intentional mis-set ≈ 3 Hz, on ISO 8608 classes, metrics in $0.1\text{--}0.5$ / $0.5\text{--}2$ / $4\text{--}12.5$ Hz bands, versus a defined passive hydro-pneumatic reference. In the $4\text{--}12.5$ Hz band, body displacement under typical road input is expected to be well below 1 mm (e.g. 0.2 g at 8 Hz ≈ 0.8 mm); mechanical actuator power in that band is therefore on the order of tens of watts, and continuous thermal load is dominated by copper loss, not mechanical work. Programme planning target (not a commitment): $\geq 10\%$ minimum improvement under frozen reference; higher goals after L1. Until results exist, comfort improvement is OPEN. SIM-001 shall also report SPORT-mode effective stiffness and roll response under lateral step inputs.

3.2 Macro–Micro Force Path

Figure 2 — Force path (conceptual)

Load type	Path	Notes
Static vertical	Upper body $\rightarrow$ capture frame $\rightarrow$ passive hydraulic/pneumatic $\rightarrow$ chassis	Never credited to magnetic force
Dynamic control (COMFORT)	Lorentz (bounded) between local carriage and upper body	Carriage follows macro; coil–magnet gap per ICD
SPORT locked vertical / roll	Upper body $\rightarrow$ lock faces $\rightarrow$ lower structure	High effective stiffness; isolation stages unloaded or held
Crash / emergency	Progressive stops + structural members	Independent of closed-loop magnetic force and of SPORT lock as primary
Power-loss	Active force $\rightarrow 0$; settle on passive support (S4); leveling valves fail-closed	Software aliveness not required

Required free-body analysis (OPEN — ATFSV-FBD-001): series/parallel load path; carriage jam; link fracture; carriage drive loss; progressive-stop engagement; SPORT lock engage/disengage under each failure. These modes are mandatory FMEA entries.

4. Actuator Topology and Lorentz Screening Numbers

Topology	Role	Notes
Hybrid hydraulic (macro) + Lorentz (micro)	REFERENCE PATH (BASELINE)	Preferred; PM-bias and 15–30 mm gap rules do not apply
Low-bias PM + EM reluctance	OPTION only	Stability gate; 15–30 mm working gap; PM bias ≤ 1.5 kN
Active hydraulic only	Macro-stage candidate	May serve leveling + macro support + lock actuation

4.1 Lorentz continuous force and power (screening)

$K_m \geq 63$ N/ $\sqrt{W}$ is an **ASSUMPTION** pending FEA or supplier data. At 0.7 kN RMS and $K_m = 63$ N/ $\sqrt{W}$, copper loss per tower is $\approx (700/63)^2 \approx 123$ W, so four towers ≈ 0.5 kW — consistent with the mean-power TARGET. At peak 2.5 kN, copper loss per tower is $\approx (2500/63)^2 \approx 1.57$ kW (pulse);

four towers $\approx 6.3 \text{ kW} \approx 131 \text{ A}$ at 48 V. ICD shall size drivers, wiring and source for this pulse class. Instantaneous force $\times$ velocity at 10–20 Hz and $\pm 7.5 \text{ mm}$ is an upper-bound product and is not representative of continuous mechanical power in the 4–12.5 Hz ride band, where body displacement is sub-millimetre and mechanical power is tens of watts; continuous thermal design is driven by copper loss. In SPORT mode Lorentz continuous demand is near zero (isolation stages held).

Quantity	Screening value	Note
Peak force	$\pm 2.5 \text{ kN}$ study per tower	Higher only after proof
Continuous RMS force (planning)	$\approx 0.7 \text{ kN}$ RMS per tower	Aligned with $\leq 0.5 \text{ kW}$ mean (4 towers)
Motor constant Km	$\geq 63 \text{ N/A}$	ASSUMPTION — FEA / supplier
Mean copper loss (4 towers @ 0.7 kN RMS)	$\leq 0.5 \text{ kW}$ class TARGET	Duty and road dependent
Peak pulse copper loss (4 $\times$ 2.5 kN)	$\approx 6.3 \text{ kW}$ class; $\approx 131 \text{ A}$ at 48 V	ICD sizing
Module supply voltage	48 V nominal class	Final in ICD
Coil–magnet gap / stroke	Few mm / $\pm 7.5 \text{ mm}$	Not 15–30 mm (OPTION only)

No topology receives safety credit from an unverified force curve. Continuous RMS force, Km, mean power and winding temperature limits shall be stated in the module datasheet (ATFSV-ACT-001) after bench measurement. Replace instantaneous 10/20 Hz force $\times$ velocity screening with ATFSV-SIM-001 power time histories when available.

5. Mass Architecture, Corner Breakdown and Module Boundary

Module mass boundary: The $\leq 55 \text{ kg}$ TARGET applies to the **core tower** (local structure + Lorentz actuator + sensors + local electronics). Hydraulic cylinder/accumulator share, valves/hoses, progressive stops, lock hardware share and shared pump unit are accounted in the full corner assembly and vehicle WBS. Full isolation corner is a STUDY allocation and is not the $\leq 55 \text{ kg}$ target.

5.1 Per-corner mass breakdown (STUDY screening)

Element (per corner)	P50 kg	P90 kg	In core tower?
Local structure	18	25	Yes
Lorentz actuator + local carriage	22	30	Yes
Sensors + local electronics	8	12	Yes
Core tower subtotal	48	67	$\leq 55 \text{ kg}$ TARGET at P50; P90 OPEN
Hydraulic cylinder + accumulator share	15	22	No — corner
Valves / hoses (corner share)	5	8	No — corner
Progressive stops	8	12	No — corner
SPORT lock hardware share	6	10	No — corner (STUDY)
Pump unit share (1/4 of shared unit)	8	12	No — shared
Full corner assembly	≈ 90	≈ 131	STUDY — not $\leq 55 \text{ kg}$ target

Core tower P50 (48 kg) is inside the $\leq 55 \text{ kg}$ TARGET; P90 (67 kg) exceeds it and requires mass reduction or TARGET revision. Four core towers at P50 = 192 kg; four full corners at P50 $\approx 360 \text{ kg}$ including lock share. Shared pump unit is counted once via the 1/4 share lines. Lock hardware mass is STUDY and must be confirmed with the lock mechanism design (ATFSV-LCK-001).

State	Upper-body mass	Static load / module	Notes
Empty (ref. vehicle)	2,150–2,350 kg	$\approx 5.3\text{--}5.8 \text{ kN}$	Curb contribution
Occupied 4 $\times$ 100 kg	$\approx 2,550\text{--}2,750 \text{ kg}$	$\approx 6.3\text{--}6.8 \text{ kN}$	Primary ref. design case
Maximum study	$\approx 2,950\text{--}3,150 \text{ kg}$	$\approx 7.2\text{--}7.7 \text{ kN}$	6 occupants + baggage
Ref. App. A cabin	1,500–2,500 kg class	$\approx 3.7\text{--}6.1 \text{ kN}$ per tower	Confirm empty vs equipped
Size S / M / L	—	$\approx 3\text{--}5 / 5\text{--}7 / 7\text{--}8 \text{ kN}$	Product size classes

5.2 Module structural load cases (screening)

Load case	Basis	Per tower (approx.)	Status
Operational (3 g vertical)	3 $\times$ static class	S: $\approx 12\text{--}15$; M: $\approx 18\text{--}21$; L: $\approx 22\text{--}24 \text{ kN}$	STUDY
Crash vertical 10 g (4-tower)	10 g on design mass / 4	L-class $\approx 65 \text{ kN}$ on 2,650 kg	STUDY
Crash vertical 20 g (4-tower)	20 g on design mass / 4	L-class $\approx 130 \text{ kN}$	STUDY
3-tower redistribution (10–20 g)	Same total / 3	$\approx 87\text{--}173 \text{ kN}$ (L-class)	STUDY — critical
SPORT lock lateral / roll (screening)	Manoeuvre + uneven lock share	OPEN — ATFSV-LCK-001	OPEN
Longitudinal / lateral crash	Buyer / vehicle standard dependent	OPEN — define with ICD	OPEN
Proof / factor	Programme to define (e.g. 1.5 $\times$ limit)	TBD	OPEN

Full vehicle curb WBS and energy/brake figures are retained in **Appendix A**. Lock lateral load cases are OPEN until ATFSV-LCK-001 freezes geometry and friction/positive-lock capacity.

6. Tower Module Architecture (ATFSV-MOD-001) — Path A Primary Product

The isolation tower set is the principal licensable subsystem. A complete Path A package typically includes: four core towers (or pair set), local controllers, ICD (mechanical / electrical / communication / safety), verification plan, defined power-loss and progressive-stop behaviour, and dual-mode SPORT/COMFORT lock interfaces. Buyer responsibility includes chassis hard-points, upper-body mass properties within the stated envelope, lock receiver structure, and vehicle-level functional safety integration.

ATFSV-MOD-001 item	Preliminary specification
Static vertical load (size class)	S $\approx 3\text{--}5$; M $\approx 5\text{--}7$; L $\approx 7\text{--}8 \text{ kN}$ per tower (screening)
Peak controlled force	$\pm 2.5 \text{ kN}$ /tower baseline; $\pm 5 \text{ kN}$ /tower study option
Continuous RMS force	$\approx 0.7 \text{ kN}$ RMS planning; OPEN thermal (ATFSV-ACT-001)
Km / mean power / supply	$\geq 63 \text{ N/A}$ ASSUMPTION; $\leq 0.5 \text{ kW}$ mean copper (4 towers); 48 V class

ATFSV-MOD-001 item	Preliminary specification
Macro / micro stroke	±50 mm (100 mm range) / ±7.5 mm; coil–magnet gap few mm (ICD)
Control bandwidth (micro)	≥20 Hz target; carriage first mode ≥3× bandwidth
Core tower mass	≤55 kg TARGET (structure + Lorentz + sensors + local electronics)
Full corner assembly mass	STUDY ≈90 kg P50 / ≈131 kg P90 (includes hydraulic + stops + lock share + pump share)
Leveling	Required; fail-closed valves; energy & failure modes in FMEA
SPORT lock	Required for dual-mode product; positive engagement; status sensors; see §22
Passive / rate-limited hold	Optional secondary (hydraulic orifice or friction); not primary crash path; failure modes in FMEA
Life	≥10,000 h equivalent road duty TARGET
Safety	Passive load support + progressive stop + independent capture + mode-transition integrity
ICD deliverable	Mechanical, electrical, communication, safety, lock ICDs
Environment (planning)	Operating temperature range to be frozen (REQ-M11); dust/water ICD

7. Module Requirements (owned) — ATFSV-MOD-001

REQ-ID	Requirement	Owner	Verification	Gate
REQ-M01	Static load capacity per size class (S/M/L) as frozen	Structures	Static bench	P1
REQ-M01b	Operational 3 g and crash load cases per §5.2 table	Structures	FEA + component test	P2
REQ-M02	Macro stroke ±50 mm without clash	Mechanical	CAD + bench	P0
REQ-M03	Lorentz stroke ±7.5 mm; coil–magnet gap per ICD	Actuator	Gap/stroke calibration	P1
REQ-M04	Bandwidth ≥20 Hz; carriage mode ≥3× bandwidth	Control	Bench dynamics	P1
REQ-M05	Peak ±2.5 kN; RMS continuous defined; Km stated	Actuator	Force + thermal	P1
REQ-M06	Core tower mass ≤55 kg TARGET	Module systems	Weighing	P1
REQ-M07	Power-loss → passive support; leveling valves fail-closed	Safety	Power-cut test	P1
REQ-M08	Active leveling preserves dynamic stroke under Δpayload	Hydraulics	Payload change test	P1
REQ-M09	Progressive stops absorb crash energy without magnet loop	Structures	Impact / FEA	P2
REQ-M10	Life ≥10,000 h equivalent road duty	Reliability	Duty-cycle / accel. test	P2
REQ-M11	Operating temperature range to be frozen (programme)	Thermal	Thermal chamber	P1
REQ-M12	EMI / magnetic exposure within limits; cabin field plan	EMC	EMC test plan	P2
REQ-M13	ICD (mech/elec/comm/safety/lock) released	Systems	ICD review	P0
REQ-M14	No single failure causes loss of geometric capture	Safety	FMEA + FTA	P1
REQ-M15	Manual recovery / rescue access provisions	Safety / HF	Access trial	P2
REQ-M16	If hold fitted: rate-limited; not primary crash path	Safety	Design review + FMEA	P1
REQ-M17	SPORT lock engages with positive confirmation; ≤3 s transition TARGET	Lock systems	Bench + vehicle test	P1
REQ-M18	COMFORT mode provides full macro envelope without lock interference	Mechanical	Stroke + lock clear test	P1
REQ-M19	Mode arbitration (driver / radar / GPS) with defined priority and override	Controls	HIL + road test	P2
REQ-M20	Transition never enters unconstrained free-float	Safety	FMEA + transition test	P1

8. Safety State Machine S0–S8 — States and Transitions

Figure 3 — State overview

State	Name	Key rule
S0	Safe / Captured	No powered release
S1	Precharge / Self-test	Capture confirmed; 48 V module supply enable path checked
S2	Normal Isolation (COMFORT)	6-DOF control; loss of capture forbidden; COMFORT active
S2S	SPORT Locked	Upper body seated and locks confirmed; isolation stages held
S3	Degraded Isolation	Reduced authority; unbounded force forbidden
S4	Power-loss Capture	Passive support; leveling valves fail-closed; free-floating forbidden
S5	Impact Preparation	Prepare stops; high-rate hold inhibited; prefer lock if available
S6	Impact / Arrest	Progressive stops; no dependence on closed-loop magnet force
S7	Post-event Safe	Supply isolation, access, rescue; no automatic re-enable
S8	Maintenance Lockout	Technician physical lockout

8.1 Transition table (PRELIMINARY thresholds)

From→To	Physical precondition	Software / logic	Timeout / note
S0→S1	Capture sensors OK; 48 V supply ready	Self-test pass	TBD s
S1→S2	Capture confirmed; stroke/gap in range; no fault	Operator enable; default COMFORT	TBD s
S2→S2S	Height at seat band; locks free; no severe fault	SPORT request (any of 3 paths)	≤3 s TARGET
S2S→S2	Locks released; stroke free; capture OK	COMFORT request	≤3 s TARGET
S2→S3	Single sensor/actuator fault or reduced authority	Degrade command	Immediate
S3→S2	Fault cleared; capture re-confirmed	Recovery enable	Via health check
S2/S3/S2S→S4	Power loss detected; valves fail-closed	Not required (hardware path)	Hardware-dominant

From→To	Physical precondition	Software / logic	Timeout / note
S2→S5	Impact prediction / severe event flag	Prepare stops; inhibit high-rate unlock	PRELIMINARY
S5→S2	False prediction cancelled; conditions normal	Cancel prep; re-enable	PRELIMINARY
S5→S6	Progressive stop engagement	Arrest logic	Event-driven
S6→S7	Motion settled; supply isolation commanded	Post-event safe	No auto re-enable
S7→S0	Manual clear; physical inspection complete	Return to safe	Manual only
S7→S1	As S7→S0 then self-test path	Via S0/S1	Manual only
Any→S8	Physical lockout device engaged	Maintenance mode	Manual only
S8→S0	Lockout removed; inspection complete	Return to safe	Manual only
S4→S1	Power restored; capture re-confirmed; sensors healthy	Self-test required before S2	Forbidden direct S4→S2
S1→S2 (after S4)	Self-test pass after power restore	Explicit operator enable	Only path back to S2

Forbidden transitions (examples): S4→S2 without S1 self-test; any state → free-floating; S6 depending on closed-loop magnetic force for energy absorption; S2S→S2 with locks still engaged. Module supply is 48 V class; vehicle HV (if any) is outside the module boundary (Appendix A). Thresholds are PRELIMINARY.

Figure 4 — Power-loss path (BASELINE invariant)

On loss of module 48 V supply (or intentional isolation): active Lorentz force → 0 → system enters S4 → macro stage settles on passive hydraulic/pneumatic support within the design envelope → leveling valves fail-closed (height retained) → progressive stops limit residual motion. If SPORT locks were engaged, they remain as mechanical constraint unless intentionally released by a fail-safe path defined in ATFSV-LCK-001. Software aliveness is not required. Return to S2 only via S1 (self-test) after power restore and explicit operator enable.

9. Summary FMEA (expanded)

Failure mode	Effect	Detection / mitigation	Sev	Occ	Det
Carriage jam	Loss of micro authority	Gap/stroke sensor; stops; passive support	9	3	4
Link fracture	Local unconstrained motion	Capture frame; stops	9	2	5
Carriage drive loss	Gap regulation degraded	Fallback passive; S3	7	3	4
Leveling stuck / delayed	Stroke consumed under payload change	Level sensor; S3	8	3	3
Leveling leakage	Slow loss of set height	Pressure/height monitor; limp	6	4	3
Hose leak / burst	Loss of hydraulic support/leveling	Pressure sensor; progressive stop	9	3	3
Accumulator precharge loss	Reduced stiffness / height authority	Pressure monitor; maintenance	6	3	4
Stop wear	Altered energy absorption	Inspection interval; design margin	7	3	5
Unwanted hold engagement	Restricted motion / false prep	Rate limit; plausibility; S3	7	2	4
Gap contamination (mud/ice)	Sensor / force error	Plausibility; cleaning; bounds	7	4	4
Coil overheating	Force derate or shutdown	Temp sensor; thermal model; threshold TBD	7	3	3
Magnet thermal demagnetisation (Lorentz)	Km reduction (force constant loss)	Temp limits on coil/magnet; derate map	7	2	4
Software / OTA fault	Incorrect command	Voting; force bounds; S3/S4	8	3	4
Unintended height drop at speed	Reduced clearance / stability	Fail-closed valves; height monitor	8	2	3
Complete power loss	Loss of active force & leveling drive	S4 passive path (invariant)	—	—	—
Sensor (stroke/force) fault	Incorrect force command	Voting; S3; force bound	8	3	3
EMI / magnetic interference	Spurious command	Shielding; filtering; bounds	7	3	4
Single tower structural failure	Load redistribution	Capture frame; remaining capacity	9	2	5
SPORT lock fails to engage	Remains in COMFORT; handling not stiffened	Lock status sensors; driver alert; stay S2	6	3	2
SPORT lock fails to release	Cannot enter COMFORT; ride harsh	Release sensor; limp SPORT; service	5	3	2
Partial lock (1–2 of 4)	Asymmetric stiffness; yaw coupling	Per-lock status; inhibit S2S if incomplete	8	2	2
False SPORT from radar/GPS	Unwanted stiffening on rough road	Driver override; confidence thresholds	4	3	3
False COMFORT in high lateral demand	Excessive roll in manoeuvre	Lateral-g / yaw rate inhibit; driver SPORT	7	2	3

Severity/Occurrence/Detection are screening 1–10 labels. Full HARA with ASIL candidates and FTA for uncontrolled separation and for partial-lock asymmetry are required at P1. Leveling valves shall fail-closed on power loss. Magnet demagnetisation on the Lorentz path reduces Km (force constant), not a bias force. Coil and magnet temperature thresholds are OPEN (REQ-M11). SPORT lock rows are new in R4.7.

10. Preliminary HARA (screening — PRELIMINARY)

Hazard	Scenario	S/E/C (cand.)	Safety goal (draft)
Loss of geometric capture	Carriage/link failure at speed	S3 / E3 / C3	Maintain mechanical capture under single failure
Uncontrolled vertical motion	Force command runaway	S2 / E3 / C2	Bound force; enter S3/S4
Loss of static support	Hydraulic burst + stop failure	S3 / E2 / C3	Progressive stops carry residual load
Unexpected height drop	Leveling valve open at speed	S2 / E3 / C2	Fail-closed valves; height monitor

Hazard	Scenario	S/E/C (cand.)	Safety goal (draft)
Inability to isolate after power loss	Passive path outside envelope	S2 / E2 / C2	S4 passive path within design envelope
Incorrect post-crash behaviour	Auto re-enable after impact	S2 / E2 / C2	S7 no automatic re-enable
Cabin magnetic exposure	Fault current / proximity	S1 / E2 / C1	Field limits + measurement plan
Maintenance access hazard	Powered motion during service	S2 / E1 / C2	S8 physical lockout required
Asymmetric SPORT lock	Partial lock under lateral load	S2 / E3 / C2	Inhibit S2S unless all locks confirmed
Unwanted mode at high lateral-g	COMFORT during emergency swerve	S2 / E3 / C2	Lateral-g / yaw inhibit of COMFORT request

S/E/C candidates are PRELIMINARY screening labels only (not formal ISO 26262 ASIL assignment). Formal HARA and ASIL determination are P1 work products.

11. Crash and Emergency Load Paths (screening)

Load case	Screening basis	Primary path
Vertical emergency (occupied)	10–20 g on design mass → total / 4 per tower	Progressive stops + upper-body structure
Braking	Peak tyre force + pitch transfer	Chassis hard-points + towers
Rollover	Roof/side contact + asymmetric tower load	Upper-body shell + capture frame
Single-tower loss	Remaining three towers with redistribution	Mechanical capture + chassis ring
Sensor/control loss	No software safety credit	Passive support + stops
Post-crash	Supply isolation / access / rescue	Independent isolation + physical access
SPORT lock during crash	Lock may share load but is not primary energy path	Stops + structure remain primary

12. Functional Safety, Cyber, Environment, Access and Service

Functional safety (planning). E/E elements of the isolation control and mode arbitration are candidates for ISO 26262 analysis. Mechanical capture, passive support and progressive stops are independent of E/E and form the primary safety case backbone. Summary HARA (§10) and FTA for uncontrolled separation and asymmetric lock are required at P1. ASIL candidates are OPEN until formal HARA is complete.

Cyber / OTA (planning). Any software update path that can change force bounds, state-machine thresholds, mode arbitration or leveling behaviour shall be subject to access control, authenticity and integrity measures. Uncontrolled OTA shall not disable mechanical capture, passive support, or force an unlock under high lateral demand without driver awareness.

Environment. Operating temperature, dust, water and chemical exposure limits are to be frozen in REQ-M11 and the ICD. Cabin magnetic field exposure shall have a measurement plan (REQ-M12) before any occupant-proximity claim. Coil and magnet temperature thresholds support demagnetisation and copper-loss derating. Lock faces and seals must tolerate road contaminants (ATFSV-LCK-001).

Access and rescue. Driver and rescue personnel retain physical access. Manual recovery provisions and post-event supply isolation are independent of the isolation control loop (S7). Access trial is a P2 gate (REQ-M15).

Service and end-of-life. Inspection intervals for progressive stops, hydraulic integrity, lock faces and sensor health shall be defined with the ICD. End-of-life handling of magnets, hydraulic fluid and electronics follows applicable regional rules — OPEN until supplier selection.

Type approval / speed. Module Path A does not by itself constitute a type-approved vehicle. Speed and homologation limits are vehicle- and jurisdiction-dependent (Appendix A notes 250 km/h as gated development only).

12.1 Standards and references (planning list)

Domain	Candidate references (verify current edition)
Ride / comfort	ISO 2631-1; ISO 8608 (road profiles)
Functional safety (E/E)	ISO 26262 (as applicable to module E/E scope)
Ambulance (if App. A)	EN 1789; EN 1865; NFPA 1917 / CAAS GVS (applicability confirmed by buyer)
EMC / magnetic	Applicable regional EMC; cabin field measurement plan
Environmental	To be frozen with ICD (temperature, IP rating)

Edition dates and exact clauses are OPEN. This list is not a compliance claim.

13. Dual Commercial Paths, Timeline and Reference Application A

Path A (primary): Isolation Module ATFSV-MOD-001 — field license or supply, including dual-mode SPORT/COMFORT capability. **Path B:** Full vehicle partnership (Appendix A). Path A does not require Appendix A powertrain or GVW figures.

13.1 Unified timeline (Path A) — contingent on funding

Phase	Month	Deliverable	Acceptance criterion
P0 — Kinematics & ICD freeze	0–2	3-D envelope, hard-stops, leveling, lock faces, ICD draft	Full stroke without clash; lock clear; ICD review
P1 — Single-tower bench	2–5	Force, thermal, power-loss, leveling, lock cycle	Force curve; passive landing; RMS thermal; lock engage/release
P2 — Four-tower rig	5–9	Four towers + representative mass + mode transitions	6-DOF; SPORT↔COMFORT ≤3 s; power-loss; stop impact
P3 — App. A mule (optional)	9–15	Module on donor chassis	ISO 2631-1 + handling comparison; structured evaluation

NDA evaluation after P0. Meaningful Path A negotiation after P1. Full module licence discussions typically require P2. Dates assume committed budget; without funding the schedule is not valid. Target funded P0 to first serious Path A discussion: 6–12 months.

13.2 Reference Application A — screening

Lead application proposals remain flexible: specialty/defence utility, luxury limited-series, or patient-transport cabin classes. Cabin mass class 1,500–2,500 kg (if used) is a screening band; confirm with the builder. Size classes S/M/L allow one module family to address multiple static-load bands. Dual-mode SPORT/COMFORT applies across these applications.

Item	Planning statement
Cabin / upper mass (if App. A)	1,500–2,500 kg class — confirm empty vs equipped with builder
Static load per tower	≈3.7–6.1 kN (4-tower) → size S/M for lighter cabins; L for heavier
Success metric (planning)	COMFORT: reduced vibration vs passive reference; SPORT: handling comparable to rigid benchmark
Candidate standards	Application-dependent (buyer confirms)
Road spectrum	Freeze with ATFSV-SIM-001 for both modes

14. Evidence Ladder (planning)

L0 Concept / architecture (this document). L1 Simulation ATFSV-SIM-001 (including SPORT vs COMFORT). L2 Single-tower bench (force, thermal, lock cycle). L3 Four-tower integrated rig (mode transitions). L4 Vehicle mule evaluation. Safety or comfort credit requires the corresponding evidence level; no level is claimed as complete in R4.7.

15. Key Technical Risks (owned) — Scale: Likelihood 1–5 × Impact 1–5

ID	Risk	L	I	L×I	Owner
R-01	Core tower mass exceeds ≤55 kg at P90	4	4	16	Module systems
R-02	Km ASSUMPTION not achieved (thermal / magnetics)	3	5	15	Actuator
R-03	Force-gap or thermal limits of Lorentz stage	3	5	15	Actuator
R-04	Macro–micro kinematic clash or carriage failure	3	4	12	Mechanical
R-05	Stroke consumed by payload without adequate leveling	3	4	12	Hydraulics
R-06	Regulatory path for novel isolation unclear	3	4	12	Homologation
R-07	Ref. vehicle P90 mass shortfall (App. A)	4	4	16	Mass systems
R-08	Tyre/brake thermal at high speed (App. A path)	4	4	16	Chassis
R-09	Comfort value unproven until L1/L2 evidence	4	4	16	Sim + PoC
R-10	EMI / magnetic exposure limits or shielding cost	2	4	8	EMC
R-11	Hydraulic mass higher than corner STUDY	3	3	9	Hydraulics
R-12	Liability if safety case incomplete	2	5	10	Legal + safety
R-13	SPORT lock reliability / partial engagement	3	4	12	Lock systems
R-14	Mode arbitration false positive/negative	3	3	9	Controls

16. Disclosure Tiers and NDA Rules

Tier	Content	Condition
T0	2-page non-enabling teaser (module name, dual-mode value, Path A offer)	Public or first contact
T1	This external baseline (module architecture, dual-mode, invariants, REQ, FMEA, PoC plan)	NDA
T2	ICD drafts, force-gap test plan, lock detail, simulation assumptions, detailed WBS	NDA + need-to-know
T3	Claim charts, design-around, pricing strategy, internal risk scores	NDA + counsel + restricted list

Detailed claim charts and prior-art analysis remain in the controlled ATFSV-IP-001 file (T3). Architecture themes are described at system level only and are not asserted patent claims.

17. Architecture Focus Themes

The following themes describe the architectural focus of the programme for technical understanding. They are not asserted claims and do not constitute a patent disclosure. Detailed claim charts remain in T3.

Theme	Description
Macro–micro kinematic split	Large body travel on passive/compliant stage with local high-bandwidth force stage
Fail-safe capture under power loss	Transition to mechanical/hydraulic support without software aliveness; fail-closed leveling
Progressive-stop energy path	Ultimate crash energy absorption independent of closed-loop magnetic force
Leveling + isolation co-architecture	Active leveling that preserves dynamic isolation stroke under payload change
Modular tower ICD + safety behaviour	Interface-controlled isolation module with defined force, stroke, bandwidth and fail-safe rules
State-machine physical preconditions	Deterministic transitions requiring physical confirmation before recovery or high-force enable
Dual-mode SPORT / COMFORT	Rigid seated lock for handling vs full isolation for soft ride; three-path selection

18. Explicit Non-Claims

This document does **not** claim or assert: (1) any measured ride-comfort improvement percentage; (2) any certification, type-approval or regulatory compliance; (3) any crashworthiness or survivability demonstration; (4) any patent grant or freedom-to-operate; (5) any production-ready mass, cost or schedule commitment; (6) any continuous public-road speed capability at 250 km/h; (7) any closed mass or energy budget suitable for production release; (8) any clinical outcome in ambulance use; (9) any continuous RMS force capability until thermal data exist; (10) any supplier or OEM endorsement; (11) any verified Km or continuous power figure; (12) any moose-test or handling ranking versus production SUVs until test data exist. All numerical targets remain subject to verification. Buyers and partners must perform independent due diligence.

19. Immediate Engineering Work Package

Pri	Work package	Closure deliverable
P0	Freeze macro–micro CAD + leveling + fail-closed valves + lock faces	3-D envelope + hard-stop + lock + service-loop drawings
P0	Release ATFSV-MOD-001 datasheet + ICD draft including lock ICD	Supplier-ready module specification
P0	Execute ATFSV-SIM-001 per §24 skeleton (COMFORT + SPORT)	Plot pack + SIM-C1...C5 review note

Pri	Work package	Closure deliverable
P0	Close core tower mass; confirm Km ASSUMPTION path; lock mass STUDY	Mass statement + FEA/supplier plan
P1	Single-tower force + thermal + power-loss + leveling + lock cycle test	Bench data package
P1	Formal HARA + FTA (capture + asymmetric lock)	Safety work products
P1	Provisional IP filing package (counsel-led)	Filing set
P2	Four-tower rig with mode transitions (P2 PoC)	6-DOF + SPORT↔COMFORT + capture + stop demo
P2	Mode arbitration HIL (driver / radar / GPS stubs)	Arbitration test report
P2	Regulatory gap assessment for target applications	Jurisdiction matrix

20. Glossary (selected)

Term	Meaning in this document
CMFSB	Captured Mechanical Floating Safety Body — not free levitation
Core tower	Local structure + Lorentz + sensors + local electronics (≤55 kg TARGET)
Full corner assembly	Core tower + hydraulic share + stops + lock share + pump share (STUDY)
Size S / M / L	Static load classes ≈3–5 / 5–7 / 7–8 kN per tower
Km	Motor force constant (N/√W); ASSUMPTION until FEA/supplier
Fail-closed	Leveling valves close on power loss to retain height
SPORT mode	Upper body seated and locked to lower body for handling (S2S)
COMFORT mode	Full macro–micro isolation for soft ride (S2)
Path A / Path B	Module field license (primary) / full-vehicle partnership (Appendix A)
S0–S8 / S2S	Safety state machine; S2S = SPORT locked sub-state

21. Master Position Statement

R4.7 is the current controlled external baseline for the ATFSV-MOD-001 isolation module. It defines a mechanically captured macro–micro isolation architecture with hybrid hydraulic macro support, Lorentz micro force control, active leveling with fail-closed valves, size classes S/M/L, consistent continuous-force and power screening numbers, a per-corner mass breakdown including lock share, owned requirements (including dual-mode), expanded FMEA, preliminary HARA, a complete state-transition table including S2S (SPORT), and explicit dual operating modes COMFORT and SPORT with three-path selection. R4.7 adds a Path A Commercial Option framework (§23) and the ATFSV-SIM-001 simulation skeleton (§24) so buyers can evaluate under a defined commercial and evidence path. Vehicle-level mass, energy and powertrain figures remain in Appendix A for Path B context only. TRL remains 2–3. No test result, certification or patentability is asserted. Founder and legal rights holder: Amirbahador Atai.

The remaining engineering task is evidence: force and thermal data, lock cycle data, L1 simulation for both modes, single-tower and four-tower demonstrations, and provisional IP filings under counsel. The architecture is sufficiently defined to support supplier engagement and controlled external disclosure under NDA.

22. Dual-Mode Architecture — COMFORT and SPORT (R4.7 Primary Addition)

R4.7 introduces a dual operating-mode architecture under familiar product names **COMFORT mode** and **SPORT mode**. The purpose is to resolve the classical conflict between soft ride isolation and high handling stiffness without requiring free levitation of the full vehicle mass. Both modes remain inside the CMFSB invariants: mechanical capture is never lost; static load is never credited solely to magnetic force; progressive stops remain the ultimate crash path.

22.1 Mode definitions

Mode	Product name	Mechanical state	Primary objective
COMFORT	Comfort mode	Locks released; macro ±50 mm available; micro active; leveling active	Soft ride; isolation vs road (ISO 2631-1 bands)
SPORT	Sport mode	Upper body driven to seat band; locks engaged and confirmed; isolation stages held	Handling; upper+lower act as near-rigid body for roll/yaw

Default after successful self-test (S1→S2) is COMFORT unless the buyer specifies otherwise in the ICD. SPORT is never assumed after power-loss recovery without a new explicit request and lock confirmation.

22.2 Mode comparison matrix

Attribute	COMFORT	SPORT
Upper–lower relative motion	Within macro ±50 mm + micro ±7.5 mm	Near zero at lock faces (compliance of structure only)
Effective roll stiffness	Soft / isolation-dominated	High / structure-dominated
Lorentz continuous demand	Up to RMS planning envelope	Near zero (held)
Leveling	Active, maintains operating point	May hold height at seat band; valves fail-closed still apply
State machine label	S2	S2S
Typical use	Rough road, long cruise, comfort priority	Winding road, emergency manoeuvre, handling priority
Driver feel target	Soft, isolated cabin	Connected, precise response
Transition target time	≤3 s into COMFORT from SPORT	≤3 s into SPORT from COMFORT

22.3 Lock and release architecture (ATFSV-LCK-001 planning)

Each corner (or a defined subset of corners) provides a **lock interface** that can seat the upper-body hard-point onto a lower-body receiver and engage a positive mechanical or hydraulic lock. The lock is not the primary crash energy path; progressive stops and the capture frame remain primary. Lock design is controlled under work product ATFSV-LCK-001 (OPEN geometry).

Element	Function	Notes / status
Seat faces	Upper hard-point contacts lower receiver when macro is driven to seat band	Geometry OPEN; wear and contamination design required
Positive lock	Prevents separation once seated (pin, claw, hydraulic clamp, or equivalent)	Must report engage/release status per corner
Actuation	Hydraulic or electromechanical drive of seat and lock	May share macro hydraulic power; duty cycle STUDY
Status sensors	Per-lock engaged / released / fault	Required for S2S entry; incomplete set inhibits SPORT
Release path	Commanded release under COMFORT request when lateral-g / yaw allow	Inhibit release under high lateral demand (threshold OPEN)
Fail behaviour	Fail-to-known: prefer remain locked or remain unlocked with driver alert — freeze in ICD	Never free-float as a fail state
Mass allocation	Per-corner lock share in §5.1 (STUDY 6/10 kg P50/P90)	Confirm with detailed design
Life	Engage/release cycles consistent with ≥10,000 h road duty TARGET	Cycle count OPEN until duty cycle frozen

Sequence for COMFORT → SPORT (TARGET ≤3 s under defined conditions): (1) Mode request accepted by arbitration; (2) Leveling/macro drives upper body toward seat band; (3) Seat contact detected; (4) Positive locks command engage; (5) All required locks report engaged; (6) Isolation stages held; (7) State becomes S2S; (8) Driver indication. Reverse for SPORT → COMFORT with release inhibit under high lateral demand.

22.4 Three-path mode selection and arbitration

Mode may be requested by three independent paths. Arbitration produces a single commanded mode. Driver authority is highest for explicit override; automatic paths may propose but shall not defeat a recent explicit driver command without defined timeout or second confirmation (policy OPEN — freeze with HMI).

Path	Name	Inputs (planning)	Typical action	Status
1	Driver selection	HMI switch / stalk / screen; optional steering-wheel control	Explicit SPORT or COMFORT; highest priority override	TARGET
2	Radar-based selection	Forward / corner radar (or lidar/camera fusion) for surface roughness, obstacles, traffic density	Propose COMFORT on detected rough surface; propose SPORT on clear high-speed or sudden obstacle field	STUDY
3	GPS / map-based selection	GNSS position + digital map road class, curvature, surface attribute if available	Propose SPORT on high-curvature map segments; COMFORT on rough-classified or urban segments	STUDY

Arbitration rule (planning)	Description
Driver explicit wins	Any explicit driver mode command overrides automatic proposals for a defined hold time (e.g. 60–300 s) or until next explicit command — policy OPEN.
Safety inhibit	High lateral acceleration or yaw rate inhibits transition into COMFORT and inhibits lock release; thresholds OPEN (freeze with vehicle dynamics).
Incomplete lock	S2S (SPORT) is not declared unless all required locks report engaged; otherwise remain COMFORT or degraded with alert.
Conflict radar vs GPS	When both automatic paths disagree, prefer the more conservative for safety (e.g. stay SPORT if either demands it under high speed) — policy OPEN.
Sensor confidence	Automatic proposals require minimum confidence; below threshold → no automatic change.
Power-loss / S4	No automatic mode change; recover only via S1 self-test and explicit enable.

Radar and GPS paths are STUDY until vehicle-level sensor suite and map licence are defined by the buyer. The module ICD shall expose mode request, mode status, lock status, and inhibit flags so the vehicle domain controller can implement arbitration without proprietary lock-in to a single sensor brand.

22.5 Transition timing and integrity requirements

Requirement	Target / rule	Status
COMFORT → SPORT complete	≤3 s under dry, level, moderate speed conditions TARGET	TARGET
SPORT → COMFORT complete	≤3 s when release not inhibited TARGET	TARGET
No free-float interval	At every sample during transition, capture and/or hydraulic constraint present	BASELINE invariant
Partial lock handling	If any required lock fails, abort SPORT entry; alert; remain COMFORT or last safe	TARGET
Lateral-g inhibit window	No unlock / no COMFORT entry above threshold (value OPEN)	OPEN
Indication	Driver receives clear mode and lock status (HMI)	TARGET
Audit log	Mode changes and inhibits logged for diagnostics	STUDY

22.6 Control and hydraulic implications

In COMFORT, macro valves and Lorentz operate for isolation and leveling as in R4.5. In SPORT, macro is driven to the seat position, locks engage, and Lorentz continuous command is reduced to hold or zero. Hydraulic energy for seating must be sized for the worst-case upper-body mass and available pump flow; simultaneous four-corner seating is the design case. Thermal benefit of SPORT is reduced copper loss; mechanical wear shifts toward lock faces and seat surfaces — inspection intervals in the ICD must reflect both duties.

22.7 Dual-mode requirements summary (cross-ref §7)

REQ-M17 (SPORT lock engage with confirmation; ≤3 s), REQ-M18 (COMFORT full envelope without lock interference), REQ-M19 (three-path arbitration with priority and override), REQ-M20 (no unconstrained free-float during transition) are mandatory for dual-mode product configurations. Single-mode (COMFORT-only) module variants may omit lock hardware but shall still meet capture and power-loss invariants.

22.8 Attractiveness and commercial note (non-binding)

Dual-mode SPORT/COMFORT addresses a known buyer objection to soft isolation systems (handling degradation in rapid manoeuvres) while preserving the isolation value proposition on rough roads. For Path A licensing, the module can be offered as COMFORT-only or dual-mode;

dual-mode is expected to broaden OEM interest. This paragraph is commercial context only and is not a performance claim.

23. Path A Commercial Option Framework (Planning — Non-Binding)

This section provides a **planning commercial structure** for Path A (module field license / supply). It is not an offer, not a price commitment, and not legal advice. Actual terms require counsel and mutual agreement. The purpose is to give Indian, Russian, Chinese and other counterparties a clear path from NDA to paid Option to license milestones, instead of an open-ended architecture discussion.

23.1 Engagement ladder (recommended sequence)

Stage	Name	Typical duration	Buyer receives	Commercial intent
0	T0 Teaser	Days	1–2 page non-enabling summary	Interest screen
1	NDA + T1	1–4 weeks	This DOC-00 (R4.7) under NDA	Technical diligence start
2	Paid Option	60–120 days	T2 package + exclusive negotiation window	Paid evaluation right
3	License / supply term sheet	After Option or PoC gate	Draft license + ICD access plan	Binding path to fees
4	Milestones + royalties / supply	Per contract	Deliverables per gate	Ongoing commercial relationship

Upfront full assignment of all IP for a single large cash payment without Option, PoC, or FTO review is not the baseline Path A structure. Buyers seeking full assignment should expect higher diligence and different pricing than a field license.

23.2 Path A Option — planning parameters (to be frozen in term sheet)

Parameter	Planning statement	Status
Option fee	Paid fee for defined evaluation period (amount negotiated; not stated as a fixed public price in this baseline)	COMMERCIAL — OPEN
Option term	60–120 days typical TARGET window from effective date	TARGET structure
Exclusivity	Optional limited exclusivity for named field / territory during Option (negotiated)	OPEN
T2 access under Option	ICD drafts, SIM-001 assumptions, lock interface detail, detailed WBS as need-to-know	BASELINE offer intent
Credit of Option fee	May credit partially against first license payment if license closes within Option + grace (negotiated)	OPEN
No license obligation	Option does not obligate either party to close a license; it funds diligence and prioritization	BASELINE principle
IP ownership during Option	Remains with rights holder (Amirbahador Atai) unless a separate assignment is signed	BASELINE
Governing law / venue	To be agreed in Option agreement (counsel)	OPEN

23.3 License / supply milestone sketch (after Option or direct term sheet)

Milestone	Gate (examples)	Typical commercial trigger
M0 — Effective license	Signed license + initial fee	Access to controlled package; field defined
M1 — ICD freeze	Buyer-accepted ICD revision	Progress payment
M2 — Single-tower evidence	Agreed bench report (force / lock / power-loss)	Progress payment
M3 — Integration support	Defined engineering support hours or design review	Service fee or included hours
M4 — Royalty / unit fee	Per module shipped or per vehicle program (if supply)	Running royalty or unit price
Minimum annual (optional)	Negotiated floor to maintain exclusivity	Use-it-or-lose exclusivity

Numeric fee levels are intentionally not published in this external baseline. They belong in a bilateral term sheet after NDA. Publishing a fixed "USD 1,000,000 upfront" in DOC-00 would be neither credible at TRL 2–3 nor flexible for different buyer sizes.

23.4 What Path A delivers vs what it does not

Under a closed Path A license (examples)	Not included unless separately agreed
Rights to use defined CMFSB module architecture in a named field	Full assignment of all related IP worldwide
Controlled technical package (ICD, datasheet, safety behaviour)	Turnkey certified vehicle or type approval
Agreed engineering support hours	Unlimited consulting or production tooling
Updates within a defined revision window (if contracted)	Open-ended R&D; for buyer's entire vehicle program
Dual-mode SPORT/COMFORT behaviour as baselined	Guaranteed comfort % or moose-test ranking without test data

23.5 Buyer actions that shorten time-to-term-sheet

Counterparties (including India / Russia / China focus) accelerate when they: (1) execute NDA quickly; (2) name a single technical owner and a commercial owner; (3) state field of use (e.g. specialty vehicle module vs full vehicle); (4) accept Option structure instead of demanding immediate full assignment; (5) fund or co-fund a minimal evidence gate (SIM-001 or single-tower). DOC-00 alone is not a substitute for those steps.

24. ATFSV-SIM-001 — Simulation Skeleton (Comfort vs Sport)

ATFSV-SIM-001 is the first evidence work product (Evidence Ladder L1). This section freezes the **skeleton**: purpose, model class, inputs, outputs, and acceptance criteria planning. It does not claim completed simulation results. Running the model and publishing numerical plots is a separate execution task (owner: simulation lead; gate: P0/P1).

24.1 Purpose and model class

Item	Planning definition
Purpose	Compare COMFORT isolation vs SPORT locked response under the same road and manoeuvre inputs; support design of macro stiffness, leveling, and lock timing
Primary model class (P0)	Quarter-car or half-car with upper-body mass, macro spring-damper (or hydro-pneumatic equivalent), optional micro force, lock as high-stiffness constraint when engaged
Extension (P1)	Four-corner model with roll DOF; lateral acceleration / step steer proxy for handling comparison

Item	Planning definition
Reference	Defined passive hydro-pneumatic corner with stated k and c (frozen before ranking claims)
Not in scope of SIM-001	Full vehicle CFD, crash FEA, magnetic FEA of Lorentz motor, detailed hydraulic CFD

24.2 Required inputs (freeze before run)

Input group	Examples	Owner
Mass	Upper-body mass (empty / occupied); corner share; sprung vs unsprung split	Mass systems
Macro stage	k, c or hydro-pneumatic curves; stroke ± 50 mm; leveling law	Hydraulics / ride
Micro stage	Force limit ± 2.5 kN; bandwidth target; or disabled in first COMFORT runs	Actuator
SPORT lock	Engaged = high k constraint; engage/release time; incomplete-lock cases	Lock systems
Road profiles	ISO 8608 class segments; at least one rough and one smooth profile	Sim lead
Manoeuvre proxy	Lateral acceleration step or simplified roll moment for SPORT vs COMFORT	Vehicle dynamics
Reference passive	Single frozen passive corner definition for all comparison plots	Sim lead

24.3 Required outputs

Output	Metric	Modes
Vertical acceleration (upper body)	RMS and peak in 0.1–0.5 / 0.5–2 / 4–12.5 Hz bands	COMFORT, reference passive
Transmissibility	A _{body} / A _{road} vs frequency (log plot)	COMFORT, reference
Stroke usage	Macro stroke histogram; time near stops	COMFORT
Roll / lateral proxy	Roll angle or lateral acceleration response to step	SPORT vs COMFORT
Lock transition	Time history of mode switch; constraint force during engage	Transition cases
Power screening	Mechanical power estimate; note copper-loss dominance for Lorentz	COMFORT

24.4 Acceptance criteria (planning — not compliance claims)

ID	Criterion (planning)	Status
SIM-C1	COMFORT shows lower RMS in at least one agreed ISO 2631-1 related band vs frozen passive reference under the same rough profile	TARGET after first valid run
SIM-C2	SPORT shows higher effective roll stiffness / lower roll angle than COMFORT under the same lateral step proxy	TARGET
SIM-C3	Macro stroke remains within ± 50 mm without continuous stop hammering on the agreed rough profile in COMFORT	TARGET
SIM-C4	Transition case does not produce unconstrained free-float interval in the model	BASELINE integrity check
SIM-C5	All plots use the same reference passive definition; assumptions listed on each figure	PROCESS mandatory

Until SIM-001 is executed and reviewed, no comfort percentage, no moose-test ranking, and no “better than OEM SUV” claim is authorized in external materials. First published result set should be marked PRELIMINARY and configuration-controlled.

24.5 Execution checklist (P0)

Step	Action	Done when
1	Freeze mass and reference passive numbers in a one-page assumptions sheet	Signed assumptions file
2	Implement quarter-car COMFORT + passive reference	Runnable model + unit tests
3	Add SPORT as locked high-k constraint	Mode switch scripted
4	Run rough + smooth profiles; export band RMS tables	Plot pack v0.1
5	Run lateral step proxy; compare roll metric	Plot pack v0.2
6	Review vs SIM-C1...C5; list gaps	Review note under NDA

Appendix A — Reference Vehicle (Path B context only)

This appendix retains vehicle-level mass, energy and brake screening figures for Path B and for sizing context. They are **not** prerequisites for Path A module licensing. Dual-mode architecture applies at module level regardless of Appendix A powertrain choices.

A.1 Curb WBS (P50 / line-sum P90) — STUDY

WBS element	P50 kg	P90 kg	Owner / note
Upper occupant/safety body structure + interior	2,150	2,350	Structures
Lower chassis + suspension structure	650	750	Chassis
ICE + accessories (1,500 hp class)	350	430	Powertrain
Generator + power electronics	220	300	Powertrain
6× motors + inverters + reductions	450	550	E-drive
Battery pack / REESS (≈ 130 kWh class)	700	850	Energy storage
HV/LV distribution + cooling	200	260	Electrical
Wheels + tyres + brakes	400	480	Chassis — high-speed rating OPEN
Isolation corners (4× full corner STUDY incl. lock)	360	524	From §5.1 ×4
Interior + HVAC + controls (vehicle share)	180	240	Cabin
Fluids + CTIS + service provisions	100	130	Vehicle
Programme mass reserve	150	200	Held outside subsystem ownership
SUBTOTAL curb (no fuel)	5,910	7,064	Aligned with corner breakdown
Fuel (usable ≈ 390 L @ 0.75 kg/L)	293	293	Energy — explicit line
TOTAL curb + fuel STUDY	6,203	7,357	Not a released weight guarantee

P90 curb+fuel + occupants can exceed 7,500 kg operational GVW. Programme options: mass reduction or raise GVW subject to tyre, brake and homologation review. Isolation corner lines follow §5.1 including lock share.

A.2 GVW configurations

Item	7,500 kg operational	6,000 kg development
Purpose	Operational accounting, range, brake, tyre, payload	Lab / mule only
Curb target (programme)	≤6,500 kg practical	≤5,000 kg objective (not closed)
Peak power / speed	3,000 hp gated; lower-speed cert path may be selected	3,000 hp / 250 km/h gated development only

A.3 Mission energy and brake (7,500 kg basis)

Quantity	Screening value	Note
Rolling power @ 120 km/h	≈29.4 kW	DERIVED
Aero power @ 120 km/h (CdA 2.3 m²)	≈52.2 kW	STUDY
Road load @ 120 km/h	≈81.6 kW	Before auxiliaries
Mission energy (1,200 km class)	≈886 kWh mechanical/road class	STUDY — ATFSV-ENE-001
Fuel with 10% reserve	≈365–391 L → TARGET ≥390 L	DERIVED
Single stop from 250 km/h	≈18.1 MJ	Kinetic
Repeated cycle 250/200/150/120	≈40 MJ cumulative class	Without regen — fade risk
Traction limit (μ≈1)	≈73.6 kN; 2.237 MW above ≈109 km/h	Speed/surface dependent

Brake thermal model (ATFSV-BRK-001) and full energy calculation chain (ATFSV-ENE-001) remain OPEN. 250 km/h is gated development only — not a continuous public-road claim.